COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / Data Structures and Analysis	**SECTION:**	CPE22S4
DATE PERFORMED:	July 29, 2026	**DATE SUBMITTED:**	July 29, 2026
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SEATWORK NO. 4.1 **Array-Based vs. STL Stack Implementations**

1. PROCEDURE OUTPUT

ILO A:

SCREENSHOT

main.cpp

```cpp
#include <iostream>
#include <stack>

int main (){
    // Create an object:
    std::stack <char> stack1;

    // Stacks: Push, Pop, isEmpty, isFull, Top

    std::cout << "Is the stack empty? " <<stack1.empty() << std::endl;
    stack1.push('S');
    std::cout << "The current top of the stack is " <<stack1.top() << std::endl;
    stack1.push('O');
    std::cout << "The current top of the stack is " << stack1.top() << std::endl;
    stack1.push('P');
    std::cout << "The current top of the stack is " << stack1.top() << std::endl;
    stack1.push('H');
    std::cout << "The current top of the stack is " <<stack1.top() << std::endl;
    stack1.push('I');
    std::cout << "The current top of the stack is " <<stack1.top() << std::endl;
    stack1.push('A');

    return 0;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> cd "c:\Users\TIPQC\Documents\DSA CARDENAS\DSA CARDENAS\HOA4.1_DeVera\" ; if
Is the stack empty? 1
The current top of the stack is S
The current top of the stack is O
The current top of the stack is P
The current top of the stack is H
The current top of the stack is I
```

OBSERVATION ANALYSIS

- This first procedure tackles the basic functions of the stacking structure. It shows us how to create a simple stacking array and how it can be a helpful programming language to use in terms of creating a library where we can implement a “first in, last out” operation.

ILO B.1

SCREENSHOT

ILOB_main.cpp ● | main.cpp

ILOB_main.cpp > Pop()

```cpp
#include <iostream>

// Global Variables:
#define maxSize 10
int stackArr [maxSize];
int top = -1, newData;

/* Function Prototypes:
    1. Top
    2. Push
    3. Pop
    4. isEmpty
    5. isFull
*/

bool isEmpty();
bool isFull();
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
========================
What number do you want to push?
3
========================
Stack operations:
[1] Push [2] Pop [3] isEmpty [4] isFull [5] Top
Enter your choice:
4
========================
Is the stack full? 0
========================
Stack operations:
[1] Push [2] Pop [3] isEmpty [4] isFull [5] Top
Enter your choice:
2
========================
Popping the top stack: 3
========================
Stack operations:
[1] Push [2] Pop [3] isEmpty [4] isFull [5] Top
Enter your choice:
5
========================
The top element is: 8
========================
```

OBSERVATION ANALYSIS	➔ In this procedure, the program is tasked to execute a stack with a maximum size of 10, as stated in the first lines of the code. It has the main functions of the stacking structures, such as push, pop, isEmpty, isFull, and top. These elements mainly displays a menu that repeatedly prompts the user to choose an operation and execute it depending of their respective functions. 1. **Push** to add an element 2. **Pop** to remove the top element 3. **isEmpty** to check if the stack has no objects 4. **isFull** to check if the stack has reached its maximum size; and, 5. **Top** to view the top element

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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    std::cout << "The current top of the stack is " << stack1.top() << std::endl;
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    std::cout << "The current top of the stack is " <<stack1.top() << std::endl;
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